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Influence of cement particles on the mechanical and buckling behavior of laminated GFRP composites with variation of end conditions of buckling

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PAPER

Influence of cement particles on the mechanical and buckling behavior of laminated GFRP composites with variation of end conditions of buckling

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Abstract

Mechanical behavior of particle-loaded laminated composites depend upon the type of matrix, filler, reinforced fibers and architecture of fibers. Further, buckling behavior of a laminated composite varies with its end conditions, i.e. free-free or fixed-fixed. Failure mechanisms like delamination, debonding and fiber fracture vary with the end conditions of buckling. Presence of hard filler may also change the buckling behavior. Thus, in the present article, the mechanical and buckling behavior of cement loaded woven glass fiber reinforced epoxy matrix composites were investigated. Cement was varied as 1%, 3% and 5% by weight of epoxy. Thickness of composites were also varied as 3.1 mm, 4.4 mm and 6.3 mm by varying the number of reinforced glass fiber mats. Tensile, Izod impact and Vickers micro-hardness tests and, buckling tests under free-free and fixed-fixed conditions were performed to assess the mechanical and buckling behavior respectively. Fractured composites were analyzed under scanning electron microscope (SEM). The results showed that tensile strength decreased and hardness increased with increase in percentage of cement. Composites with minimum thickness showed lowest tensile strength. Critical buckling load increased with increase in thickness of composites for both end conditions. However, the nature of variation of buckling load with the percentage of filler varies for every thickness. SEM images showed that damage of composites mainly involved fiber breakage and delamination. Cement particles were effective in deflecting the advancing crack.

1. Introduction

The demand of advanced engineering composites with tailorable properties for structural applications is increasing day by day [1–3]. Researchers are paying extensive efforts to develop such composites. Glass fiber reinforced polymer matrix (GFRP) composites are considered as ideal composites for structural applications due to their excellent mechanical properties and low cost [4–6]. The main limitation of GFRP composites lies in their low interface strength due to which, chances of delamination increases [2, 7]. The main factors which affect the strength of interface are chemical bonding, Vander Waals interaction and molecular segregation [7]. Three strategies are generally adopted to improve the delamination properties, i.e. (1) 3D fiber reinforcement, (2) modification of polymer matrix and, (3) combination of both. Modification of polymer matrix is generally used due to ease of fabrication of composites. Polymer matrix is generally modified by adding micro or nano-sized particles which generally lead to enhancement in mechanical properties [8–10]. Parameters like particle size, shape, type and nature of matrix/particulate bond affect the mechanical properties of polymer matrix composites [7, 11]. Studies on cement particles filled polymer matrix composites showed that incorporation of cement in the polymer matrix may enhance its mechanical properties [12, 13].

A significant number of studies are available on particulates reinforced composites, but studies on particle modified matrix material for fiber reinforced composites are limited. Particulate modified fiber reinforced polymer matrix composites are gaining attention for their use in structural applications [14]. The synergistic

effect in these composites lead to enhancement in mechanical properties. Particulates, as secondary reinforcement in the polymer matrix, are generally mixed with polymers before its impregnation in the fiber system [15]. Continuous fibers provide more strength to the composite as compared to the short fibers [16]. Thus continuous fibers are generally preferred over short fibers where strength is the main requirement. Woven fabric composites are easy to fabricate as compared to unidirectional composites. Thus woven fabric composites are generally used.

The impact damage mechanism in particulate filled fiber reinforced composites depend on the type of matrix, fiber, and filler [17]. The percentage of particulate filler influences the damage resistance of composites [18]. It can't be generalized that at a particular percentage of filler, composite will yield highest damage resistance as well as all other mechanical properties. Further, it can't be generalized that incorporation of a particular filler will lead to improvement of all mechanical properties of fiber reinforced composites [8]. The local plastic deformation and microcracking of composite material during impact affect its damage resistance [19]. The modification of matrix by a particulate filler can have a severe impact on its local plasticity and microcracking behavior [20]. Adding a chemically resistant hard particulate filler in a fiber reinforced composite may increase its hardness, impact strength and chemical resistance of the resulting composite [21].

Buckling behavior of particles filled GFRP composites varies with the percentage of filler, particle size, and nature of filler [22]. Critical buckling load varies with the percentage of filler [23]. Sayer [22] found that incorporation of hard particulate filler in a GFRP composite may lead to increase in its critical buckling load by 42%. Further, the nature of end conditions also affect the damage mechanism in buckling of FRP composites. Variation of end conditions lead to variation in delamination size and geometry which strongly affects the buckling behavior [24, 25]. The mechanism and position of delaminated layers affect the critical buckling load. Critical buckling load is strongly influenced by the length of delamination closest to the surface if near surface delaminations are longest, however, size of delaminations beneath also have significant influence on buckling load. [26] Local buckling in a sublaminates also affects the critical buckling load and it also depends on end conditions [27]. Local buckling can occur prior to the appreciable global buckling. The buckling behavior of FRP composites has been investigated mostly with both ends free or one fixed and one free end [28–30]. Buckling behavior of FRP composites with both ends fixed has not been investigated yet.

According to best of our knowledge, the mechanical and buckling behavior of cement modified epoxy matrix impregnated GFRP composites has not been investigated yet. Thus in the present article, cement particle toughened epoxy matrix based GFRP composites were investigated for their mechanical and buckling behavior. To assess mechanical behavior tensile, Izod impact and Vickers micro-hardness tests were performed. Buckling behavior was investigated with both ends free and both ends fixed. The weight percentage of cement filler was varied. The thickness of composites was also varied.

2. Material and methods

2.1. Material

Cross-woven ($0^\circ/90^\circ$) E-glass fabrics were used for the preparation of GFRP composites. The composite specimens were prepared by hand lay-up technique. Different weight percentages of cement were added as 1%, 3% and 5% by weight of the matrix. Firstly, the required amount of epoxy (PL 411) was taken in a beaker and cement was added to it. Then the mixture was stirred mechanically for almost 30 min to get a uniformly dispersed solution. After proper mixing of the epoxy and cement, hardener (PH 861) in the ratio of 1:10 by weight of epoxy, was added to it, and the entire mixture was again mixed. During mixing, temperature increased in the range of $65^\circ-70^\circ$ due to cross-linking of polymer chains which indicated the beginning of reaction process. When the mixture started heating, it was applied layer by layer on fiber lamina, i.e. an alternate layer of lamina and epoxy was applied to prepare the laminated composites. Pressure of 8 KPa was applied on the top face, to enhance infiltration of epoxy so that neither epoxy accumulation occurs nor dry patches of the fibers occur. The composites were left for curing in compressed condition for about 18 h. After curing, composites were cut to prepare samples for different mechanical tests.

2.2. Sample preparation

Fabricated composite plates were cut to obtain the samples for tensile test, Izod impact test, Vickers micro-hardness test and buckling test. For tensile test, samples were prepared according to the ASTM D3039 standard. For Izod impact test, samples were prepared according to ASTM D256 standard with different thickness. For micro-hardness testing, rectangular specimens having dimensions 25 mm $\times$ 25 mm were cut from the composite plate.

For buckling tests, composite plates were cut in the dimensions of 240 mm $\times$ 20 mm $\times$ t with thickness ranging 3.1 mm, 4.4 mm and 6.3 mm. All the prepared specimens were polished on the sides using emery paper

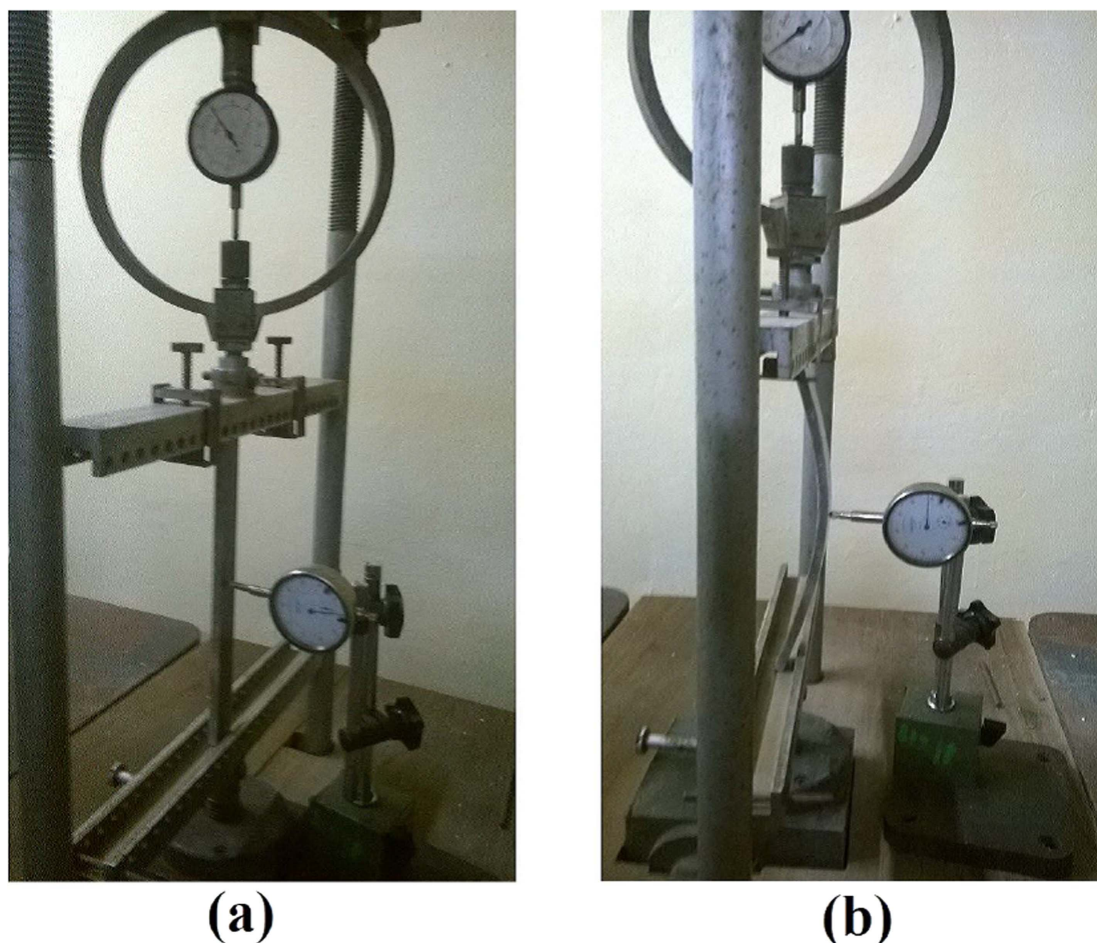


Figure 1. (a) Buckling set-up at the start of the test and (b) Buckled mode shape of the specimen during testing.

having grades 50, 200 and 600, to attain the dimensional accuracy and surface finish as well. End surfaces were flattened to prevent movement or slipping of the sample while testing. The specimens were cut using wire cutting machine at high speed to prevent delamination [31].

2.3. Mechanical tests

Tensile tests were performed at room temperature. The speed of moving head was kept maintained at 2 mm min^{-1} . Vickers microhardness tests were performed to determine the microhardness of the composite samples. Microhardness testing was carried out on the Vickers microhardness tester (Model: Semi-Automatic, Sr No. 104, Micro Mach Technologies, Pune, India). The hardness tester was provided with a diamond indenter in the form of a right pyramid of a square base of an angle 136° between opposite faces. To eliminate any agglomerate effect, a total of five readings were taken for every sample at different locations on it. A load of 500 g was applied with a dwell time of 30 s.

Izod impact tests were performed on an impact test equipment (Model; Resil Impactor-50, CEAST, S.p.A., Italy). The impact length and velocity used were 0.327 m and 3.426 m s^{-1} . The mass of hammer that was dropped onto the specimen was 0.932 kg. The impact strength was obtained as per ASTM D256. The specimens were fixed in the testing machine such that the notch root lies in level with the vice and the hammer strikes on the notched face.

Buckling tests were performed to determine the critical buckling load for the composite specimens. Buckling was done under two end conditions which were (i) both ends free and (ii) both ends fixed. It was analyzed during the buckling tests that in case of both ends free, all samples got fractured from the middle of the span while in case of both ends fixed, some samples got fractured from the clamped ends. Buckling set up at the start of test and during testing in case of both ends free, is shown in figure 1.

2.4. Scanning electron microscopy

After performing tests, the samples fractured under Izod impact tests and buckling tests were analyzed under scanning electron microscope (SEM, Serial no.- EVO 18-20- 45, ZEISS EVO 18 RESEARCH, Germany).

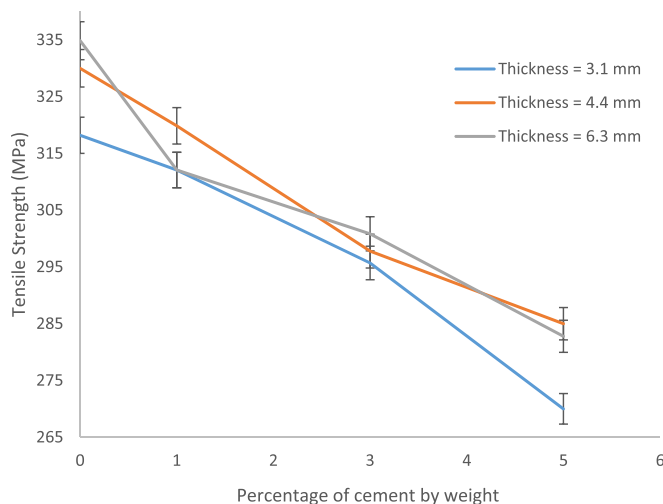


Figure 2. Variation of tensile strength of GFRP composites with cement content (wt%).

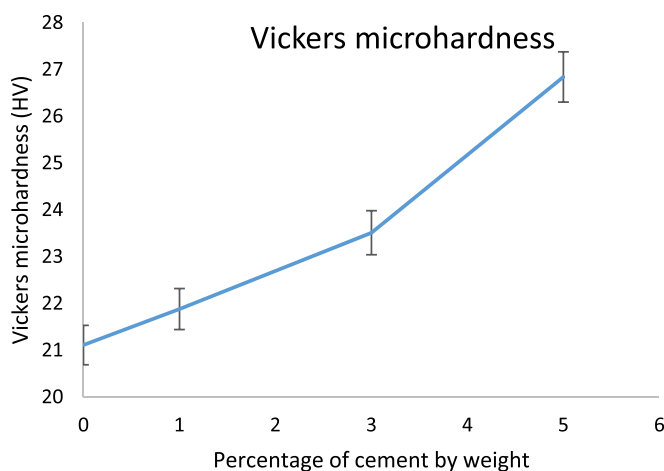


Figure 3. Variation of microhardness with cement content (wt%).

3. Results

3.1. Effect of filler content and thickness on tensile strength of GFRP composites

Figure 2 shows the tensile strength of cement modified GFRP composites. It can be observed from figure 2 that tensile strength decreased with increase in weight percentage of cement.

The composites with 3.1 mm thickness showed lowest tensile strength at all percentages of filler. Composites showed highest tensile strength when epoxy was not modified with cement particles.

3.2. Effect of filler content on microhardness of GFRP composites

Figure 3 indicates that the Vickers micro hardness increased with increase in the percentage of cement. The increase was sharp when the cement content was more than 3%.

3.3. Effect of filler content on Impact strength of GFRP composites

It can be observed from figure 4 that as the weight percentage of cement in the composite was increased, the energy consumed for the fracture of specimen first increased, then decreased and again increased after that. Composites with 3.1 mm thickness absorbed lowest impact energy.

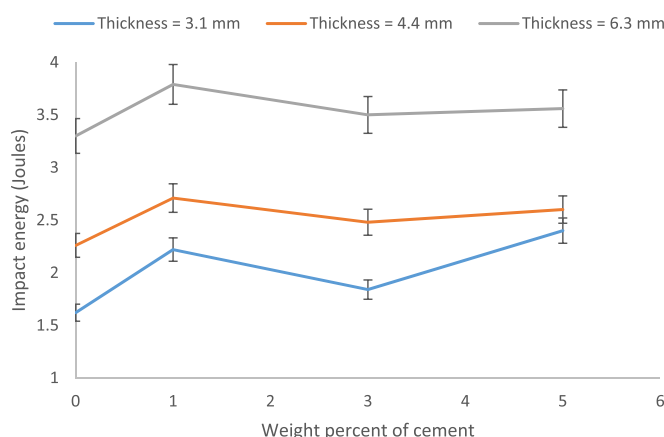


Figure 4. Variation of Impact energy absorbed versus cement content (wt%).

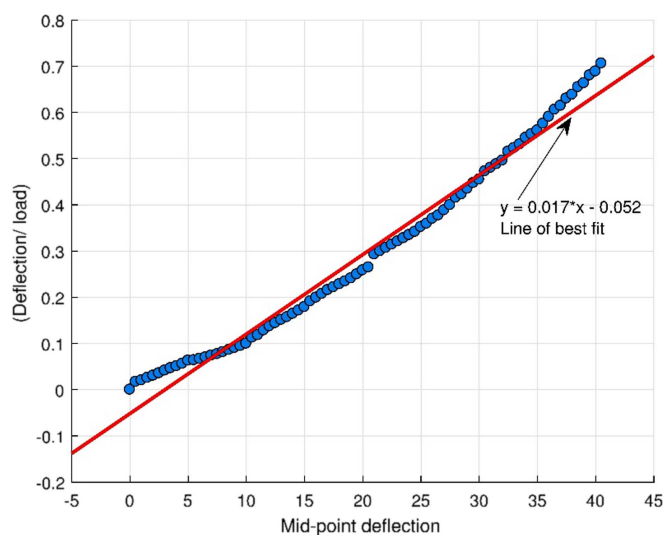


Figure 5. Southwell plot for GFRP composite specimen having thickness of 3.1 mm.

3.4. Buckling behavior of cement filled GFRP composites with both ends free

Figure 5 shows the Southwell plot for GFRP composite without modification of epoxy matrix when both ends were free. Southwell plot is used to determine the critical load capacity of the material. Inverse of best fit line's slope gives the critical load.

Figure 6 shows the representative plots for buckling behavior of cement loaded GFRP composites when both were free. It can be observed from the figure that GFRP composite with 5 wt% of cement realized the highest value of critical buckling load when the thickness was 3.1 mm.

Figure 7 shows the variation of critical load capacity of unfilled and filled E-glass/epoxy composite when both ends were free. The critical load value was highest for 5% cement filled specimen in all thickness conditions. But in each case, at certain weight fraction, there was a dip in the critical load value. The critical load value decreased and was minimum for 1% filled specimen and after that it increased for 3% and 5% filling and yielded maximum at 5%.

Figure 8 shows the representative plots for buckling behavior of cement loaded GFRP composites when both were fixed. It can be observed from the figure that GFRP composite with 1 wt% of cement realized the highest value of critical buckling load when the thickness was 3.1 mm.

Figure 9 shows the variation of critical load capacity of unfilled and filled E-glass/epoxy composite when both ends were fixed. The critical load value was minimum for unfilled, 1% and 3% cement content in 3.1 mm, 4.4 mm and 6.3 mm thick composites respectively.

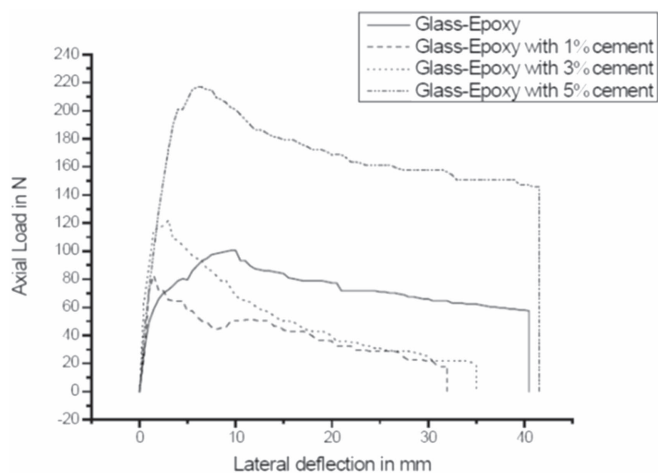


Figure 6. Lateral deflection versus axial load for specimen thickness 3.1 mm with both ends free.

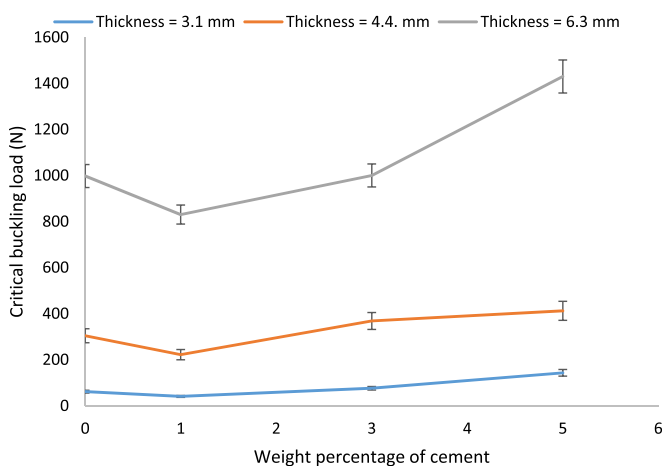


Figure 7. Variation of critical buckling load with cement content when both ends were free.

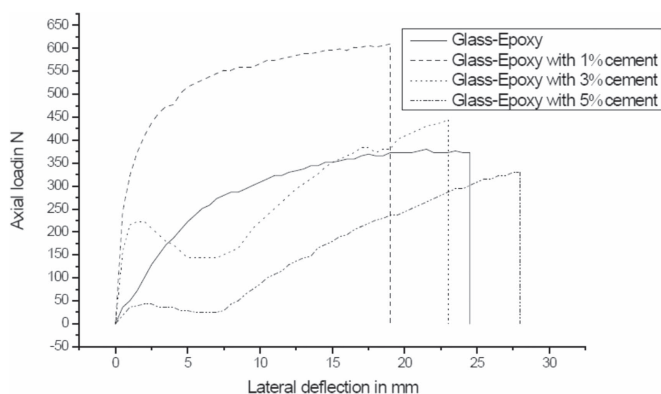


Figure 8. Lateral deflection versus axial load for specimen thickness 3.1 mm with both ends fixed.

4. Discussion

It can be observed from figure 2 that tensile strength of woven fabric GFRP composites decreased with increase in percentage of cement filler, no matter what the thickness was. This was attributed to the accumulation of cement particles near fiber/matrix interphase region. The cement particles which were accumulated nearby

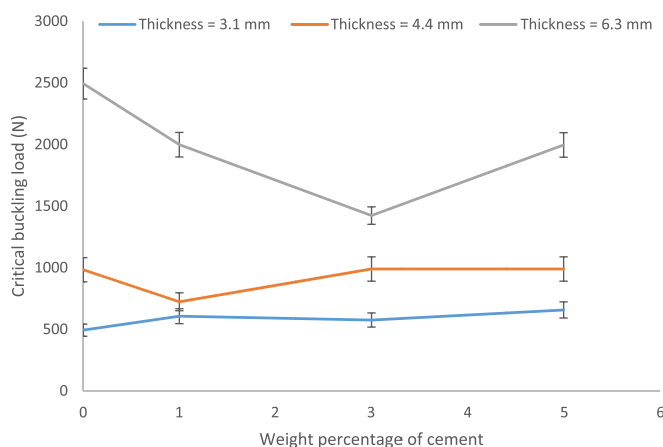


Figure 9. Variation of Critical buckling load with cement content when both ends were fixed.

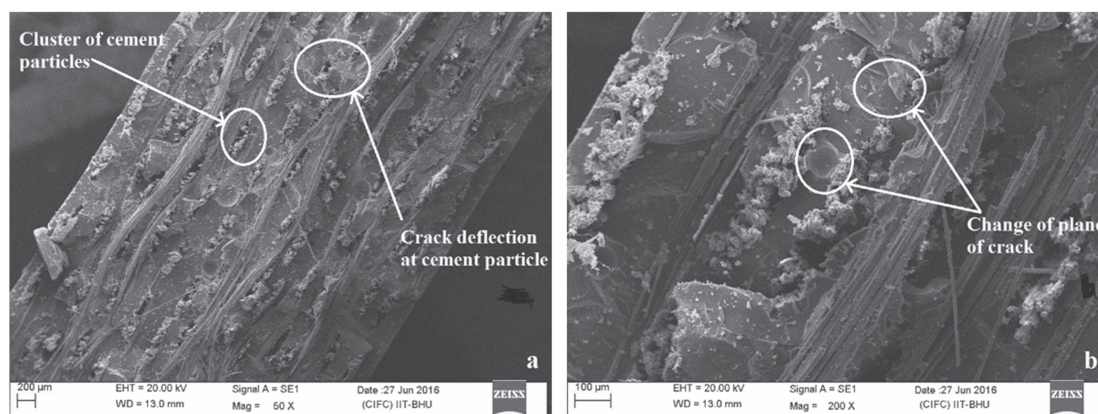


Figure 10. Specimen (1% cement, 3.1 mm thick) fractured under Izod impact test.

fiber/matrix interphase, decrease strength due to weak fiber/matrix interphase bonding. [32] Tensile strength also varied with the thickness of the composites due to advancement/delay in the onset of crack initiation due to thickness effect [33]. Microhardness increased with increase in the percentage of cement, as can be observed from figure 3. This may be attributed to the increased resistance to localized deformation due to incorporation of hard particulates.

It was observed (figure 4) that impact strength of GFRP composite increased for 1%, decreased at 3% and again increased for 5% of cement by weight. If there is little interfacial adhesion, particulate fillers increase the brittleness of ductile matrices. The reverse is true for brittle matrices. [34] figure 10(a) shows the SEM image of the fracture surface of 1% cement reinforced GFRP sample after impact. It can be observed that the crack changed its plane and direction at some locations near cement particles. Maximum regions in the composites were covered with uniform dispersion of cement particles but some clusters can also be observed. Fiber/matrix interface was strong enough so that, at the time of fracture, fiber strands came out from the fiber region. This increased the impact strength of the composite. It can be observed from figure 10(b) that in matrix, advancing crack propagated through different planes which consumed energy and thus required more energy to propagate. As the percentage of cement was increased, agglomeration of cement particles led to decrease in impact energy.

As the number of plies increased, the crack propagated through the fiber/matrix interface as shown in figure 11(a). This required more energy for fracture of the composite. Thus impact energy increased as number of plies in the composite increased. Matrix fracture with deep cracks can be observed in figure 11(b).

Figure 7 shows that when both ends were free, critical buckling load first decreased up to 1% filler and increased after that. This may attributed to the poor infiltration of filler in between the different plies which reduced the overall pinning effect and thereby decreased the buckling strength. Moreover, the cohesion force between epoxy is more as compared to cement/matrix interface strength. The synergism between overall crack pinning effect and weak interface strength led to decrease in buckling strength upto 1% cement loading. As the

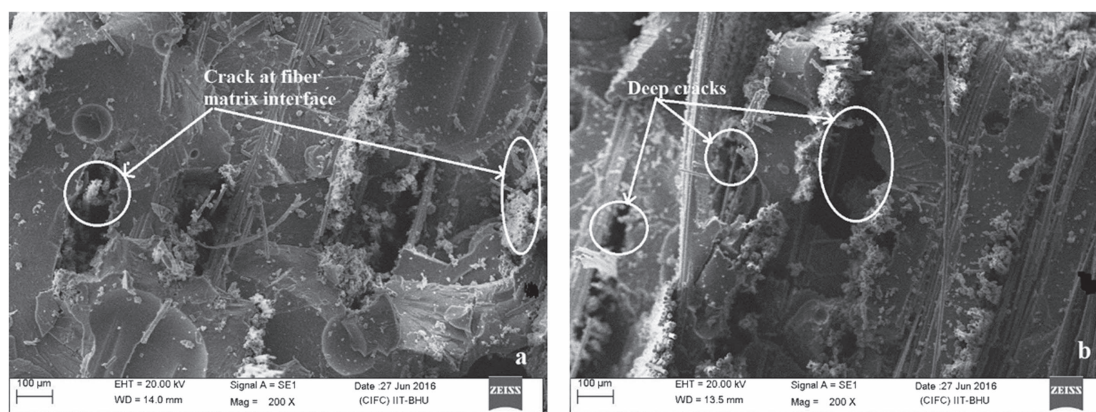


Figure 11. Specimen (1% cement, 4.4 mm thick) fractured under Izod impact test.

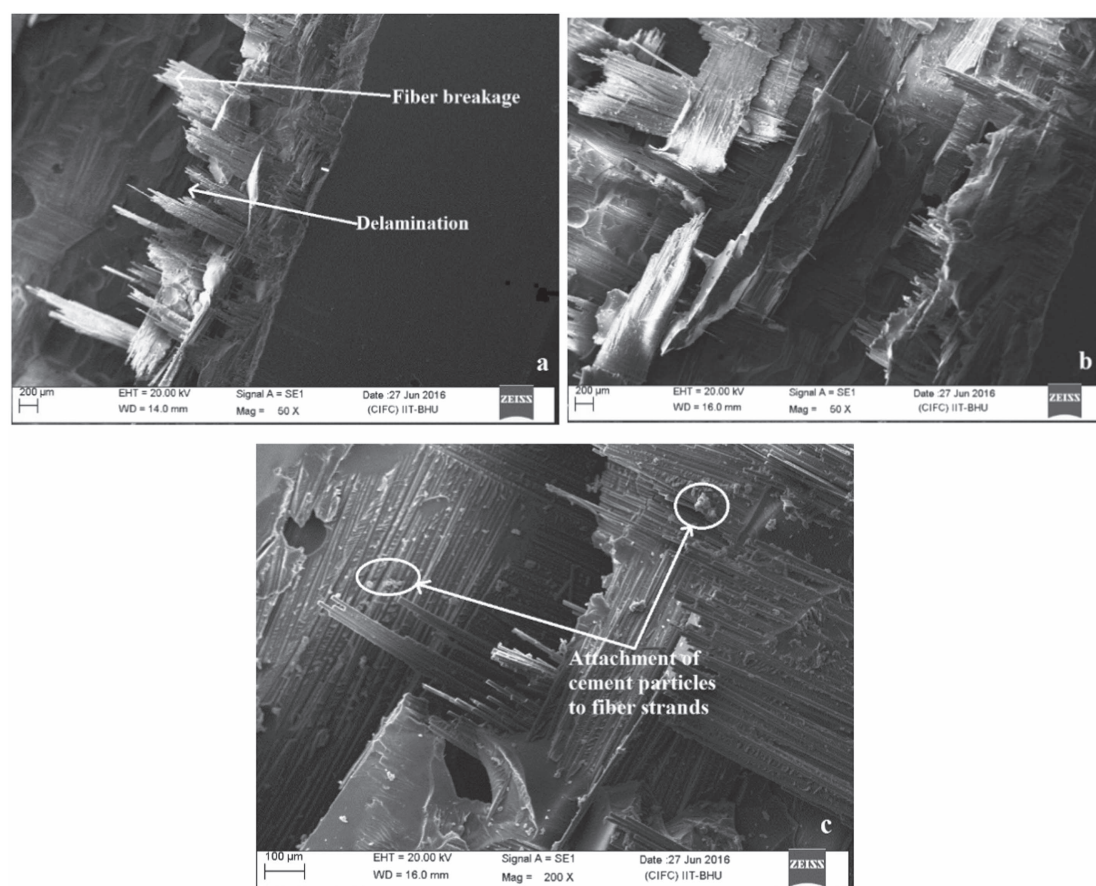


Figure 12. Fracture surface of sample (1% cement, 3.1 mm thick) failed in buckling with both ends free.

percentage of cement increased, cement particles were well dispersed in between the plies of the composite which led to increase in buckling strength.

It can be observed from figure 12(a) that in case of GFRP composites having 3.1 mm thickness, fiber breakage with delamination fracture took place. However, fracture took place at an angle to the longitudinal axis of the composite as observed in figure 12(b). There were some regions where the matrix material came out of the composite from other planes as observed in figure 12(b). It can be observed from figure, 12(c) that some cement particles got attached to the fiber strands and changed the direction of crack during its propagation. Thus, as the percentage of cement was increased, critical buckling load generally increased.

When the composites were loaded with both ends free, there was smooth transition of load from the ends to the center of the specimen. Allowing rotation at both the ends made the composite to take the shape of half sine

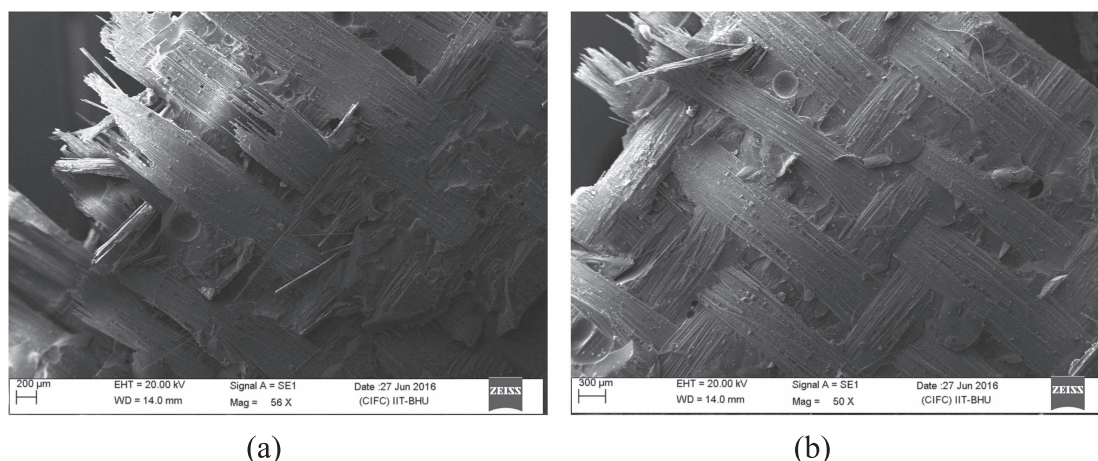


Figure 13. Fracture surface of sample (1% cement, 6.3 mm thick) failed in buckling with both ends fixed.

wave which led to increase the tensile stresses at the outer surface of the composite laminate. Crack initiated at the outer surface of the laminate led to delamination of outer surface ply. The plies were free to relax themselves after delamination. Incorporation of cement particles tends to change the direction of crack propagation. Thus the plane of delamination for different plies were different. It requires more energy to change the plane of crack propagation which led to increase in critical buckling load with an increase in percentage of cement.

When both ends were fixed, the fixed ends were not free to rotate. Thus transverse deflection mainly occurred near mid-span of the composite. The curvature formed due to transverse deflection was not smooth which reduced the chance of delamination and fiber breakage occurred as can be observed in figure 13. However, fiber breakage didn't occur in a single plane. Fracture plane was different for different plies. The fracture surface was irregular and serrated in appearance. Buckling strength didn't show any specific trend when both ends were fixed. Critical buckling load increased with increase in thickness due to increase in load sharing capacity with increase in number of plies. The synergism between effective crack pinning, filler dispersion due to thickness effect and how smoothly the load was transferred, governed the buckling behavior of cement loaded GFRP composites.

It was observed that critical buckling load was less when both ends were free as compared to both ends fixed conditions. This may be attributed to the state of stress and the deflection at the center line of the specimen. In case of free ends, the ends were free to rotate, thus led to large deflection at low loads. This generated high stress at the outer surfaces of composites nearby center line. However, when the ends were fixed, centerline deflection was somewhat restricted because ends were not free to rotate. Thus, high load was required for a significant centerline deflection. This increased critical buckling load.

5. Conclusion

The effect of cement particle reinforcement on mechanical and buckling behavior of GFRP composite was investigated in this study. Variation in end conditions of buckling was also considered in the present investigation. Based on the results it was inferred that:

- Tensile strength decreased with increase in cement content whereas hardness increased.
- Impact energy increased with increase in the number of plies and its minimum was observed at 3% cement content due to agglomeration or non-uniform dissolution of cement particles in resin.
- In case of buckling with both ends free, critical buckling load increased with increase in number of plies when the cement content was more than 1%. Cement particles led to effective crack pinning.
- Transverse deflection at the center line changes with the change in end conditions. Free movement of ends in free ends condition led to the fracture of composites at low buckling load as compared to fixed ends condition.
- The synergism between crack pinning, filler dispersion and how smoothly the load was transferred governed the buckling behavior of cement loaded GFRP composites.

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